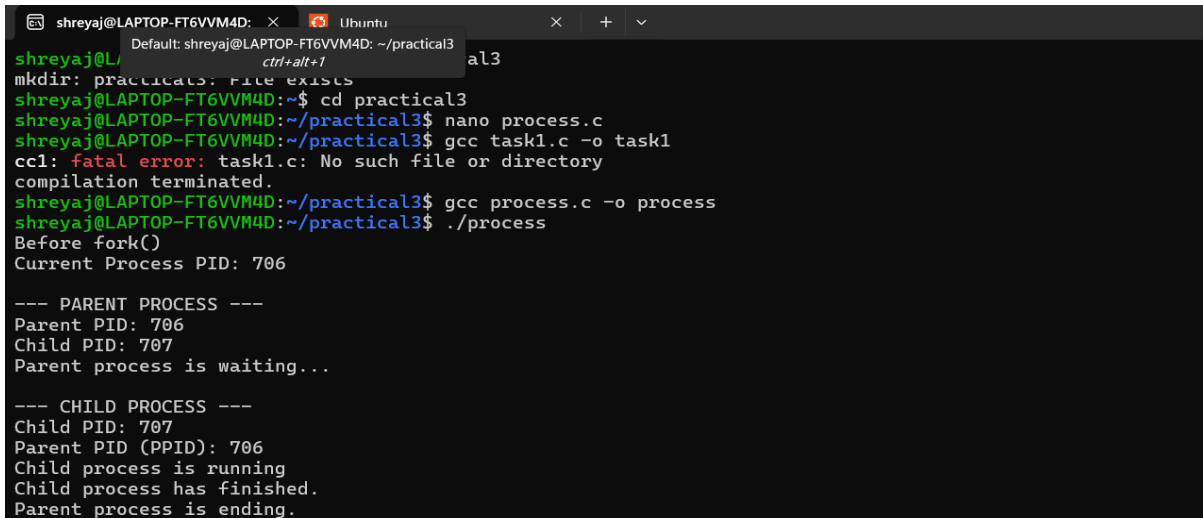


PRACTICAL WEEK - 3

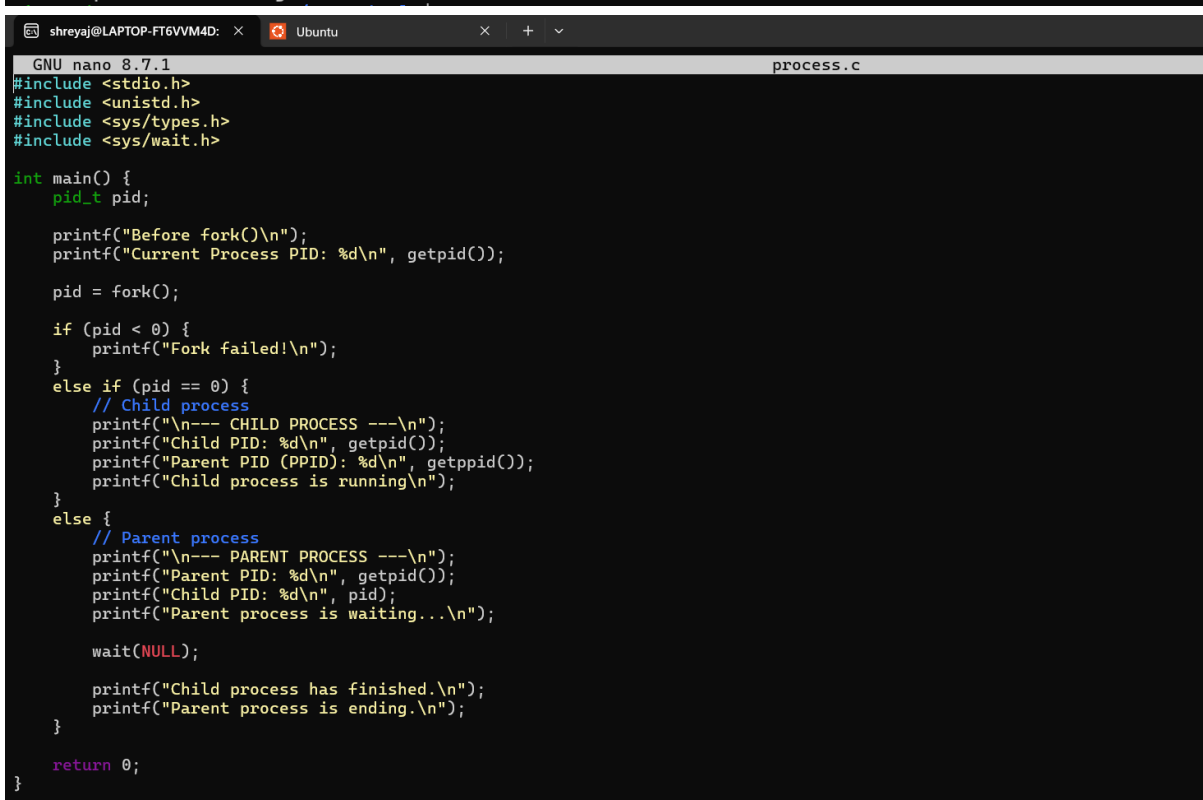
Task -1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.



```
shreyaj@LAPTOP-FT6VVM4D: ~$ cd practical3
shreyaj@LAPTOP-FT6VVM4D: ~/practical3$ nano process.c
shreyaj@LAPTOP-FT6VVM4D: ~/practical3$ gcc task1.c -o task1
cc1: fatal error: task1.c: No such file or directory
compilation terminated.
shreyaj@LAPTOP-FT6VVM4D: ~/practical3$ gcc process.c -o process
shreyaj@LAPTOP-FT6VVM4D: ~/practical3$ ./process
Before fork()
Current Process PID: 706

--- PARENT PROCESS ---
Parent PID: 706
Child PID: 707
Parent process is waiting...

--- CHILD PROCESS ---
Child PID: 707
Parent PID (PPID): 706
Child process is running
Child process has finished.
Parent process is ending.
```



```
GNU nano 8.7.1 process.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Before fork()\n");
    printf("Current Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID (PPID): %d\n", getppid());
        printf("Child process is running\n");
    }
    else {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent process is waiting...\n");

        wait(NULL);

        printf("Child process has finished.\n");
        printf("Parent process is ending.\n");
    }

    return 0;
}
```

TASK – 2 : Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
shreyaj@LAPTOP-FT6VVM4D:~/practical3$ nano process.c
shreyaj@LAPTOP-FT6VVM4D:~/practical3$ gcc process.c -o process
shreyaj@LAPTOP-FT6VVM4D:~/practical3$ ./process
Before fork()
Current Process PID: 1092

--- PARENT PROCESS ---
Parent PID: 1092
Child PID: 1093
Parent process is waiting...

--- CHILD PROCESS ---
Child PID: 1093
Parent PID (PPID): 1092
Child process is running
Child process has finished.
Parent process is ending.
```

```
shreyaj@LAPTOP-FT6VVM4D:~$ ps -el
F S  UID  PID  PPID  C  PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S   0     1      0  0  80   0 -  6144 -   ?           ?    00:00:00 system
5 S   0     2      1  0  80   0 -   795 -   hvc0        ?    00:00:00 init-s
0 S   0     7      2  0  80   0 -   795 -   hvc0        ?    00:00:00 init
4 S   0    47      1  0  79  -1 - 12600 -   ?           ?    00:00:00 system
4 S  989    79      1  0  80   0 -   5604 -   ?           ?    00:00:00 system
4 S   0    90      1  0  80   0 -   8845 -   ?           ?    00:00:01 system
4 S   0   113      1  0  80   0 -    722 -   ?           ?    00:00:00 chrony
4 S   0   120      1  0  80   0 -   1118 -   ?           ?    00:00:00 cron
4 S  995   124      1  0  80   0 -   2205 -   ?           ?    00:00:00 dbus-d
4 S   0   157      1  0  80   0 -  11024 -   ?           ?    00:00:00 networ
4 S   0   164      1  0  80   0 -   4609 -   ?           ?    00:00:00 system
4 S  100   179      1  0  80   0 -  55137 -   ?           ?    00:00:00 rsyslo
4 S  988   237     113  0  80   0 -   5106 -   ?           ?    00:00:00 chrony
1 S  988   246     237  0  80   0 -   3024 -   ?           ?    00:00:00 chrony
4 S   0   257      1  0  80   0 -  30865 -   ?           ?    00:00:00 unatte
4 S   0   282      1  0  80   0 -   1373 -   tty1        ?    00:00:00 agetty
4 S   0   329      2  0  80   0 -   2161 -   ?           ?    00:00:00 login
4 S 1000   373      1  0  80   0 -   5586 -   ?           ?    00:00:00 system
5 S 1000   377     373  0  80   0 -   5729 -   ?           ?    00:00:00 (sd-pa
4 S 1000   402     329  0  80   0 -   1534 poll_s pts/1    ?    00:00:00 bash
5 S   0   560      2  0  80   0 -    796 -   ?           ?    00:00:00 Sessio
5 S   0   561     560  0  80   0 -    800 -   ?           ?    00:00:00 Relay(
4 S 1000   562     561  0  80   0 -   1567 poll_s pts/0    ?    00:00:00 bash
5 S   0   1098      2  0  80   0 -    796 -   ?           ?    00:00:00 Sessio
5 S   0   1099    1098  0  80   0 -    800 -   ?           ?    00:00:00 Relay(
4 S 1000   1105    1099  0  80   0 -   1567 do_wai pts/2    ?    00:00:00 bash
5 S   0   1137      90  0  80   0 -   8846 -   ?           ?    00:00:00 (udev-
5 S   0   1138      90  0  80   0 -   8846 -   ?           ?    00:00:00 (udev-
0 R 1000   1144    1105  0  80   0 -   1798 -   pts/2    ?    00:00:00 ps
```

```
shreyaj@LAPTOP-FT6VVM4D: ~$ top
top - 05:52:31 up 29 min, 1 user, load average: 0.02, 0.01, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7776.6 total, 7131.4 free, 559.2 used, 237.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7217.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24576   15540  11568 S   0.0   0.2   0:00.84 systemd
    2 root        20   0    3180    2204    2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2040    1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50400   16764   15468 S   0.0   0.2   0:00.23 systemd-journal
   79 systemd+    20   0   22416   14500   12012 S   0.0   0.2   0:00.10 systemd-resolve
   90 root        20   0   35380   12176    9080 S   0.0   0.2   0:01.40 systemd-udev
  113 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.05 chronyd-starter
  120 root        20   0    4472    3176    2924 S   0.0   0.0   0:00.01 cron
  124 message+   20   0    8820    5404    4724 S   0.0   0.1   0:00.07 dbus-daemon
  157 root        20   0   44096  29800   14660 S   0.0   0.4   0:00.28 networkd-dispat
  164 root        20   0   18436    9428    8184 S   0.0   0.1   0:00.11 systemd-logind
  179 syslog     20   0   220548  5904    4612 S   0.0   0.1   0:00.07 rsyslogd
  237 _chrony     20   0   20424    6448    5668 S   0.0   0.1   0:00.09 chronyd
  246 _chrony     20   0   12096    2424    1780 S   0.0   0.0   0:00.00 chronyd
  257 root        20   0  123460  32604   17996 S   0.0   0.4   0:00.26 unattended-upgr
  282 root        20   0    5492    2756    2572 S   0.0   0.0   0:00.01 agetty
  329 root        20   0    8644    5612    4772 S   0.0   0.1   0:00.02 login
  373 shreyaj     20   0   22344   13604   10984 S   0.0   0.2   0:00.18 systemd
  377 shreyaj     20   0   22916    4076    2076 S   0.0   0.1   0:00.00 (sd-pam)
  402 shreyaj     20   0    6136    5420    3856 S   0.0   0.1   0:00.04 bash
  560 root        20   0    3184    1116     980 S   0.0   0.0   0:00.00 SessionLeader
  561 root        20   0    3200    1256    1108 S   0.0   0.0   0:00.06 Relay(562)
  562 shreyaj     20   0    6268    5540    3864 S   0.0   0.1   0:00.09 bash
 1098 root        20   0    3184    1116     980 S   0.0   0.0   0:00.00 SessionLeader
 1099 root        20   0    3200    1256    1108 S   0.0   0.0   0:00.02 Relay(1105)
 1105 shreyaj     20   0    6268    5612    3936 S   0.0   0.1   0:00.06 bash
 1156 shreyaj     20   0    8320    6108    3896 R   0.0   0.1   0:00.34 top
 1237 root        20   0   35384    6460    3332 S   0.0   0.1   0:00.00 (udev-worker)
 1238 root        20   0   35384    6460    3332 S   0.0   0.1   0:00.00 (udev-worker)
```